

Session 3

Networking Fundamentals

Understanding how devices talk to each other across a network



Lecture by

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Topics Covered

01

TCP/IP Protocol Framework

The rules that govern how data travels across a network

02

IP Address

The unique digital address assigned to every device

03

Subnet Masks

Dividing an IP address into network and host portions

04

CIDR Notation

Flexible, efficient allocation of IP address blocks

TCP/IP Protocol Framework

What is TCP/IP?

TCP/IP is the framework that defines how data moves from one device to another across networks.

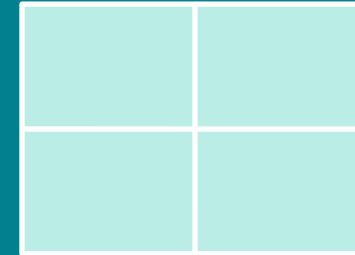
Transmission Control Protocol / Internet Protocol

It is the set of rules computers use to communicate over a network or the Internet.

Think of it like sending a parcel:

IP → Where should the parcel go?

TCP → Make sure the parcel arrives correctly and in order.



PARCEL

*IP addresses it. TCP delivers it —
complete, correct, in order.*

TCP/IP =

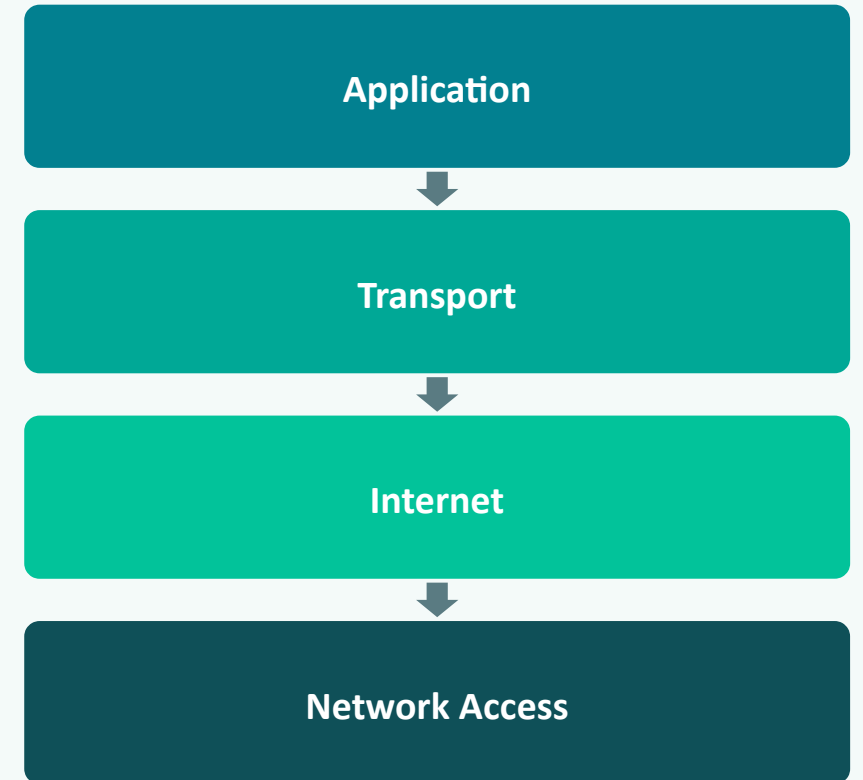
**Transmission Control Protocol /
Internet Protocol**

TCP/IP Has 4 Layers

Layer	What it does	Examples
Application	Services applications use	HTTP, DNS, FTP, SSH
Transport	End-to-end communication	TCP, UDP
Internet	Addressing & routing	IP, ICMP
Network Access	Sends data over the network	Ethernet, Wi-Fi

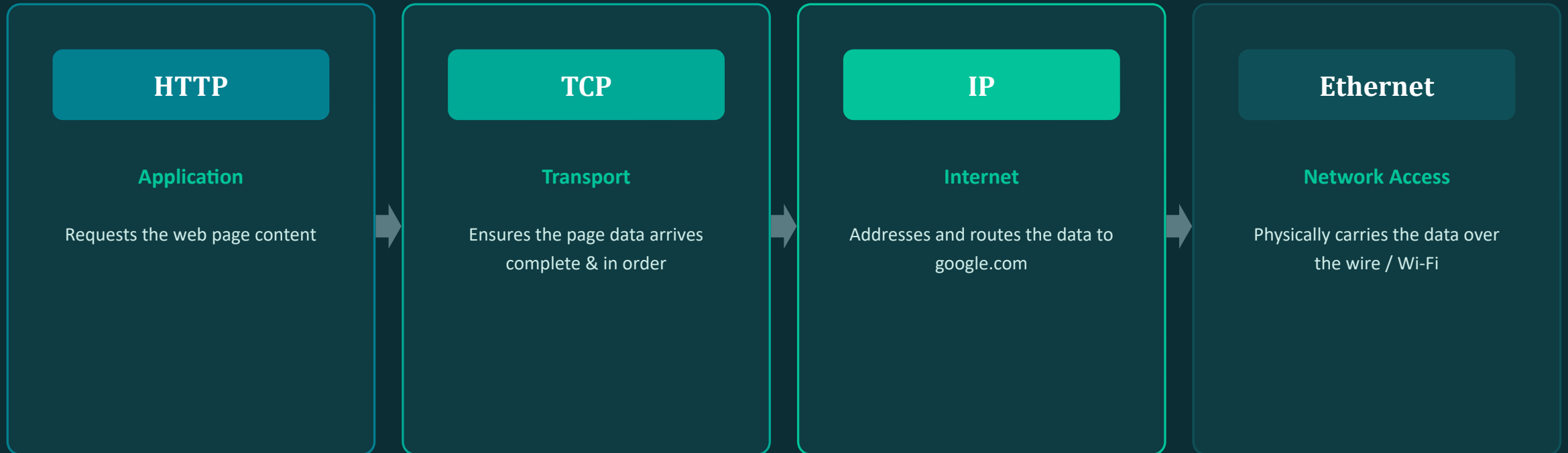
Data flows down to send, and back up to receive.

Easy way to remember



A Real Example

Opening a website walks through every layer of TCP/IP



IP Addresses

What is an IP Address?

An IP address (Internet Protocol address) is a unique label assigned to every device connected to a computer network.

It acts like a digital home address, allowing devices to find and talk to each other.



192.168.1.10

Its unique digital address

Every device on a network — phones, laptops, servers
— has one.

Types of IP Addresses

Public IP

Assigned by your internet provider to your entire home or office network. Visible to the whole internet and allows you to access websites.

Private IP

Assigned by your router to individual devices (phone, laptop) inside your local network. Keeps devices safe by hiding them from the public internet.

Dynamic IP

Temporary addresses that change automatically over time. Your internet provider rotates them to save resources — most household devices use these.

Static IP

Permanent addresses that never change. Manually set for servers, business networks, or websites needing a constant, reliable connection.

IPv4 vs IPv6

IPv4

32

bit address

4.3 Billion

unique addresses

Format example

192.168.1.1

Disadvantage

Limited number of addresses (only 4.3 billion); not secure.

IPv6

128

bit address

340 Undecillion

unique addresses

Format example

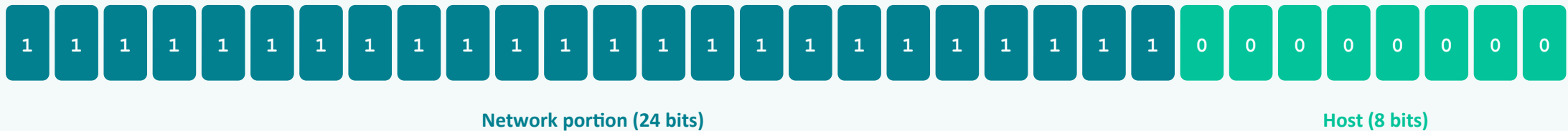
2001:db8::1

Disadvantage

Long, hard-to-remember address format.

Subnet Masks

A **subnet mask** is a 32-bit number used with an IP address to divide it into two parts: the **network ID** and the **host ID**.



Binary 1's → represent the network portion of the IP address.

Binary 0's → represent the host portion.

The subnet mask helps a device determine whether another device is on the same local network or a different network — deciding whether communication is direct or must go through a router.

Classless Inter-Domain Routing (CIDR)

CIDR is a method of IP address allocation and routing that allows more efficient use of IP addresses — allocating based on a network **prefix** rather than a fixed class (A, B, or C).

Why CIDR? — Classful addressing wastes IP addresses

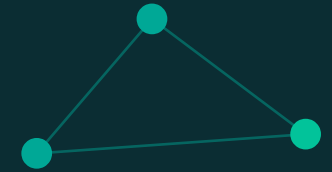
Class	IPs Available	Hosts	Example Wastage
A	2^{24}	$2^{24} - 2$	Too large for small orgs
B	2^{16}	$2^{16} - 2$	Wastes 49,150 hosts for 214 needed
C	2^8	$2^8 - 2$	Small networks only

Problem

Organizations often need a host count that doesn't match class sizes — leading to wastage.

Solution

CIDR allows flexible block allocation matching the exact number of addresses an organization requires — eliminating the fixed-class wastage.



Thank You

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